Software-Projektpraktikum (SPP)
Supervised Process Intelligence

Project Setup

Group 1

Remaining Time

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1 Introduction

A remaining time prediction system for an incident management process can assist in detecting and consequentially alerting process workers to a potentially problematic process instance, as well as supporting resource arrangement decisions (e.g. whether to add more workers to aid the execution of a certain activity to meet the deadline) [1], which can further satify the goal of an incident management, namely "identify and correct problems while maintaining normal service and minimizing impact to the business" [4]. Since this project focuses on the prediction of a running system, one highlight is to provide real-time feedback to the process workers, enabling immediate and flexible adjustment.

2 Project Value

Remaining time prediction can be described as one of the *most ambitious* forms of process mining [2], which can bring value to both other operational support activity of the system and its users with more intuitive view in the performance of certain activities. Moreover, remaining time prediction can also help carry out a “de jure model" from the “de facto model".

2.1 Process's Side

2.1.1 Diagnostics

As for the process model derived from the given event log, remaining time prediction can also be annotated on whose representation (e.g. process map) and serve as diagnostics for reference, which enhances the process model with data perspective.

Based on such diagnostics, certain visualization methods can be performed to provide more intuitive insight in the process for end users (e.g. bold-marked outgoing arcs from a place with high remaining time in a Petri-net, or deeply colored box of the activity in such condition).

2.1.2 Recommend

Recommendations, which is also an operational support activity, can be made on predictions [2]. In the frame of this project, the remaining time of a running incident is selected as the performance indicator.

For instance, if multiple prediction values can be provided, along with their occurrence probability, recommendations on the best next activity to execute can be made in terms

of the shortest remaining time. Other types of recommendations, like the best resource allocation be to adopted, or with which label the current running case should be archived later (e.g. “normal case" or “strongly delayed case").

2.2 Organization’s Side

As for an organization, the remaining time of a running incident is a common Key Performance Indicator (KPI), because it can be associated with other KPIs, (e.g. cost, resource allocation/utilization etc).

2.2.1 Resource Utilization and Efficiency

If a relatively precise remaining time prediction can be made, resource utilization can be probably improved, since the waiting time between execution of instances is possible to be shortened - which further reduces the total cost, enhances the workflow execution efficiency and even the Service Level Agreements (SLA) compliance in general [2].

2.2.2 Adjustment/Intervention

Moreover, if certain adjustment/intervention must be applied (e.g. overly long resource occupation expected, or expected failure in meeting the deadline), remaining time prediction can also provide numeric details for decision making (e.g. resource reallocation, rearrangement of activity execution sequence, waiting queue restriction) or serve as reference for other analysis.

2.2.3 Loss/Damage/Violation Estimation

Even if no more further remedies could be carried out, the prediction can also contribute to loss/damage estimation (e.g. help find out an optimized action to take in order to minimize the loss) or consequences/intensity of possible violations [3].

2.2.4 Alert and Bottleneck Detection

Furthermore, if a certain activity is observed to be predicted with relatively long remaining time frequently, alert to the process worker about potential bottlenecks can also be generated. In more severe cases, the collection of such alert can still indirectly support the process redesign.

2.3 User's Side

As mentioned in the last subsection, remaining time is a relatively common KPI for an organization, the output of this project is a numeric value which does not necessarily have to be interpreted by professional process analysts, but can also be directly offered to end users [2].

User-friendliness, especially for users who are less affiliated with process mining, is as well a highlight of this project - which can, on the other hand, share the organization certain cost for staff training. Moreover, such real-time prediction can improve users' experience by providing interaction with the monitored case and its context.

3 Theoretical Feasibility Study

In the context of process mining, prediction of the remaining time of a running case (alias incident) can not be performed unconditionally. In [2], it is noted that prediction is only possible for *Lasagna processes* (i.e. a *structured process* [1], where it's possible to create an agree-upon process that has a fitness of at least 0.8 with limited effort) and there must be an advanced IT infrastructure to provide high-quality event logs, which is also allowed to embed an operational support system.

In this section, we will first try to explain why those prerequisites for prediction are fulfilled in this project. Then, additional considerations beyond those rigid requirements will be listed as recommendation for the implementation.

3.1 Prerequisite Fulfillment

3.1.1 Lasagna process

The target of this project is an incident management process, which can be defined as "a process used by IT operations and DevOps teams to respond to and address unplanned events that can affect service quality or service operations" [4].

C. Alberts et al [5], have already compared on several definitions of processes within the incident management system and found out that most works include similar set of processes despite various terminologies adopted, which are: "Prepare", "Protect", "Detect", "Triage" and "Respond". Additionally, in this project, "Start" and "Complete" should also be taken into consideration, since the data source is event log.

[1] As defined in [2], a structured process is a process in which all activities are repeatable and have a well defined input and output

Alternatively, the IBM version is also recommendable [4], as “Identify the incident", “Log and classify the incident", “Contain the issue", “Diagnose the incident", “Resolve the incident" and “Close and review the incident".

In the aspect of input and output, they are also well-defined, as respectively event logs captured from historical execution and a numeric value of the remaining time of a running incident

Therefore, it is clear that the incident management process satisfies the definition of a *Lasagna process* according to [2], and prediction can consequentially be performed on such process.

3.1.2 Data Source and Infrastructure Environment

As recommended by our advisor, we will train our model primarily based on BPI Challenge 2013 Incident Management log [2], which is directly incident handling and has at least case ID, activity name and timestamp - those are the fundamental prerequisite to further predict the remaining time for this project. An example carried out from ProM 6.15 is displayed in Figure 1.

Therefore, along with other attributes and relatively sufficient amount of events as primary training data (over 65 thousand events listed), this data set can be considered as “high-quality".

If other training data sets shall be applied to this project in the future, they should also preserve the “high-quality" standard.

However, it is impossible to assure that the infrastructure environment is also satisfying. Given the relatively mature development of incident management systems/processes, like the National Incident Management System (NIMS), we can still assume that good quality of input data can be guaranteed.

3.2 Additional Considerations

Even though the baseline of the theoretical feasibility of this project has been discussed and proven, there are still other aspects that should be concerned during the implementation.

For example, in [2], it is emphasized that processes are not always in steady state. They can rather be influenced by working hours, weekends, contextual factors and drifts.

Beyond the in-born problems of event logs, context is also hard to define in the real

[2] https://www.processmining.org/event-data.html

Figure 1: Example carried out from ProM 6.15 for existence of essential event log attributes

implementation: is the context merely the “de jure model" that is learned from “post mortem" event data, or the activities that are executed in parallel and even in the same queue should be integrated? We will carry these question and try to find out whose answer on the way of trials and errors encountered in this project.

4 Tools

Trello: To manage tasks and track project progress, we use Trello with a Kanban-style. Tasks are organized into categories such as “Backlog”, “In Progress”, “In Review” and “Done”. This helps the team coordinate work, assign responsibilities, and monitor deadlines throughout the project life-cycle. In this way, we become more aware of each other and strive to work more harmoniously.

GitHub: GitHub is used as the primary platform for version control and collaborative development. Each team member works on branches, and changes are integrated through pull requests followed by code reviews. This workflow enables effective change tracking, minimizes merge conflicts, and supports a structured and transparent development process.

PM4Py: In this project, PM4Py is used to load and preprocess event log data. By presenting the functions necessary for loading and structuring logs, it ensures that each entry is properly ordered by timestamp. These prefixes are important for calculating the remaining time and preparing the data for machine learning models. Additionally, since this system includes features such as filtering, sorting, and feature selection, it will help us lay the keystones for the project. These steps must be followed to ensure clean and consistent input data. PM4Py makes it easy to convert raw event logs into structured representations; this data can then be processed using various tools.

Pandas: Pandas is a data analysis library used to manipulate and transform structured

data. It can be used to load and process event log data, especially when working with CSV files. In this project, it allows us to organize the data into a tabular format, perform efficient operations such as filtering and aggregation, and compute features required for the prediction task. These include metrics like elapsed time, number of events in a prefix, and activity-related statistics. Overall, pandas plays a key role in preparing the final dataset that is used as input for the machine learning models.

ProM: ProM can be used to validate the quality of event logs by checking important attributes such as case ID, activity and timestamp. Although it is not part of the main pipeline, ProM can support exploratory analysis and future project extensions.

Disco: Disco is a process mining tool used for analysis of event log data. In this project, it can be used to better understand the structure and behavior of the event log, which can support feature engineering and data validation steps.

Scikit-Learn: We use scikit-learn to implement and evaluate machine learning models for predicting remaining time. It provides a consistent interface for training regression models, such as linear regression and random forest. In this project, scikit-learn is used to analyse the relationship between prefix-based features and the remaining time of a case. scikit-learn also supports data splitting, preprocessing and hyperparameter adjustment.

Matplotlib: Matplotlib is a visualisation library used to create plots and graphical representations of data. In this project, it can be used mainly during the evaluation phase to analyse the performance of machine learning models. It allows us to generate plots. These include predicted vs. actual remaining time. They also include error distributions. And they allow for performance comparisons across different prefix lengths. These visualisations help us to see how well the models perform and spot any patterns or weaknesses in the predictions.

5 Team

5.1 Members

Our team consists of three Computer Science students. Each member brings relevant academic background and programming experience to the project, allowing us to effectively tackle the upcoming challenges:

- **Sarp Aydoslu:** "I am a seventh-semester Bachelor Computer Science student with a growing interest in process mining. My interest in this field developed during the Business Process Intelligence course, where I gained fundamental knowledge and

practical experience. For this project, I am particularly interested in remaining time prediction and the use of machine learning in real-time process monitoring. Although I have not yet completed a dedicated project in this area, I am motivated to expand my practical experience and deepen my understanding. I see this field as a potential direction for my future career, which is why I am eager to take part in this project. I have basic programming skills in Java, Python, and C, which I believe will help me contribute effectively."

- **Linas Butkus:** "I am a fourth-semester Bachelor's student in Computer Science with a strong interest in data science and machine learning. I bring fundamental knowledge of Machine Learning acquired through a recent proseminar, where I specifically focused on predictive models such as Decision Trees and Neural Networks. I am highly motivated to apply these theoretical concepts to real-world scenarios. This project presents an excellent opportunity to bridge the gap between algorithmic theory and practical software engineering. Additionally, I contribute solid Python programming experience, which will be valuable for the implementation of our system."
- **Qifan Wu:** "I contribute a strong theoretical foundation from knowledge gained from both the Business Process Intelligence (BPI) and Business Process Management & Computation (BPMC) courses. The mixture of my genuine interest in mathematics with process mining and BPM ignites my passion in this project. Although I'm still a rookie in software project development except the project carried out from PSP practical lab, I still believe that, with my theoretical background, I can rapidly get on the track of this project with my teammates and meanwhile deepen my understanding of 'paper-based' knowledge with practical experiences, as well as in teamworking."

5.2 Roles

We have agreed on a clear yet flexible distribution of roles. This approach is specifically designed to minimize single points of failure, ensure a fair balance of the workload, and guarantee that every team member remains fully engaged and aligned with the project's overall progress.

Furthermore, we consider it essential that documentation and reporting are not isolated tasks. Instead, all members share the responsibility of maintaining high-quality documentation and reporting throughout every development phase and within their respective domains.

Based on these principles, the primary responsibilities are distributed as follows:

- **Sarp Aydoslu:** Evaluation & Visualization Lead, Quality & Tooling (Tests, CI)
- **Linas Butkus:** Data & Preprocessing Lead, Modelling & Experimentation Lead
- **Qifan Wu:** Project Coordinator, Technical Lead

6 Project Charter & Schedule

Weekly meetings take place every Sunday to outline upcoming project steps and allocate tasks. These sessions serve as an opportunity to discuss current developments, address questions, and exchange ideas directly. Following the meeting, team members work independently on their assigned responsibilities. To ensure a well-coordinated process and a successful project outcome, this workflow is supported by a flexible communication structure and clear, punctual deadlines.

In addition to our internal schedule, we have consolidated the general project deadlines below to serve as a central reference for our time management:

Milestone	**Deadline**
Project Setup	April 24, 2026
Requirements Engineering	May 3, 2026
Sprint 1	May 17, 2026
Sprint 2	May 31, 2026
Sprint 3	June 14, 2026
Assessment Document	June 21, 2026

7 Phase Review

7.1 Reflection on the Initial Phase

Overall, the project setup phase was highly productive and allowed us to establish a strong organizational foundation. On the very evening of the official group kick-off, we proactively held our first internal team meeting. This immediate action enabled us to efficiently coordinate our workflows and set up the necessary technical infrastructure, such as our GitHub repository and Trello dashboard, marking a successful start to Milestone 0.

During this initial meeting, we transparently discussed our individual schedules, personal time constraints, and overall availability. This led to the mutual agreement on our

weekly Sunday meetings, ensuring that everyone's workload is respected while maintaining continuous project progress.

The primary challenge we currently face is a lack of deep, practical experience with the specific domain of remaining time prediction, coupled with the general time constraints typical of a university semester. However, we view this as a natural part of the learning curve and are confident that we can overcome this through a structured, step-by-step familiarization process.

7.2 Knowledge Gaps & Learning Objectives

To actively address our lack of domain-specific experience, we have identified key areas where we need to build expertise. We have distributed the learning responsibilities among the team to ensure efficient progress:

- **PM4Py Framework:** As we currently lack hands-on experience with PM4Py, all team members will be responsible for studying the documentation and familiarizing themselves with the library.
- **Testing & Python Pipelines:** We need to improve our practical knowledge regarding test code composition and automated coding pipelines in Python. All team members will actively engage in learning standard Python testing practices.
- **Prediction Theory:** The specific theoretical background required for making predictions based on event logs demands a deeper understanding. Linas and Qifan will take the lead in researching and compiling these theoretical concepts.
- **Software Architecture:** To ensure a robust and scalable build-up of the system, a clear structural design is necessary. Sarp will be in charge of researching and defining the overall software architecture for this project.

References

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